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基于变分/梯度的方法，逐个双比特门局部优化，与Decomposition of Matrix Product States into Shallow Quantum Circuits论文有点类似，都是先有一个框架，然后二次/多次优化

实现的量子态制备
目标是用尽可能少的量子门

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Approximate encoding of quantum states using shallow circuits

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Quantum algorithms and simulations often require the preparation of complex states through sequences of 2-qubit gates. For a generic quantum state, the number of required gates grows exponentially with the number of qubits, becoming unfeasible on near-term quantum devices. Here, we aim at creating an approximate encoding of the target state using a limited number of gates. As a first step, we consider a quantum state that is efficiently represented classically, such as a one-dimensional matrix product state. Using tensor network techniques, we develop and implement an efficient optimization algorithm that approaches the optimal implementation, requiring a polynomial number of iterations. We, next, consider the implementation of the proposed optimization algorithm directly on a quantum computer and overcome inherent barren plateaus by employing a local cost function. Our work offers a universal method to prepare target states using local gates and represents a significant improvement over known strategies.

Quantum state preparation is the task of generating a circuit that prepares a user-defined target state. This task is the first step of several quantum algorithms, such as linear solvers[1], quantum machine learning[2,3], and quantum recommendation systems[4], which require encoding of classical information in a quantum state. Similarly, in the quantum simulation of nonequilibrium systems, the initial state is often a complex state, for example, corresponding to the ground state of a many-body Hamiltonian[5–8]. Because a generic state of N qubits is defined uniquely by 2^N independent amplitudes, encoding such a state requires an exponentially large number of gates[9–19].

In real devices, the maximal number of gates per circuit is heavily limited by noise and decoherence. Quantum computing practitioners often assume that the maximal achievable circuit depth scales linearly with the number of qubits (see, e.g., the definition of quantum volume by IBM[20]). To match this requirement, it is necessary to heavily reduce the total number of gates used in state encoding. One possible strategy is to consider segments of the circuit, searching for a smaller circuit that performs the same operation, for example, by scanning lists of known circuit identities[21–23]. This method deterministically reduces the number of gates, but is limited to previously studied problems and does not apply to generic multi-qubit states. Another common method is to perform a blind search in a variational space of quantum circuits[24–31], trying to incrementally approach the target state. The problem with this approach is that due to the large number of variational parameters, the optimization algorithm is, again, limited to a small number of qubits only.

In this paper, we develop an efficient method to obtain an approximate encoder, given a fixed number of gates. We, first, consider quantum states that can be efficiently represented classically, such as matrix product states (MPS)[32–36]. In this case, the optimization algorithm can be realized efficiently on a classical computer, using resources that scale polynomially with the number of gates. We exemplify our protocol for both MPSs with low entanglement and random MPSs and demonstrate a significant improvement with respect to known techniques. Next, we consider the realization of our algorithm on a quantum computer, where the optimization protocol is affected by inherent shot noise. This error source leads to barren plateaus that limit the applicability of optimization algorithms. We overcome this limitation by introducing an improved version of the algorithm, based on local cost functions, whose resource requirements scale polynomially with the number of gates.

In the first part of our work we encode quantum states that can be efficiently expressed using classical means, and in particular, using open-boundary matrix-product states (MPS). In this representation, a one-dimensional state of N qubits is expressed in terms of $2N$ matrices, $\{A_i^{(0)}\}_{i=1}^N$ and $\{A_i^{(1)}\}_{i=1}^N$ according to

$$|\psi\rangle = \sum_{j_k\in\{0,1\}} A_1^{(j_1)} A_2^{(j_2)} \dots A_N^{(j_N)} |j_1 \dots j_N\rangle. \quad (1)$$

Here, $|j_1 \dots j_N\rangle$ is the computational basis and we assumed open boundary conditions, such that the first and last matrices are, respectively, row- and column- vectors. The maximal dimension of A_i is denoted by χ, and is known as the bond dimension. Here, we will consider realistic situations where the bond dimension is in the intermediate regime $1 \ll \chi \ll 2^{N/2}$. As we

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will see below, χ determines the resources required to realize the MPS on a quantum computer in an exact manner. By fixing χ, one obtains a variational family of states with maximal entanglement entropy $S_{\max} = \log_2\chi$. For this reason, MPSs offer a faithful description of the ground state of one-dimensional gapped Hamiltonians, whose entanglement entropy follows an area law and does not grow with N[37–40]. Similarly, MPS is useful in quantum machine learning tasks, such as ansatz structures[41] and ansatz pertaining[42].

From Eq. (1), it is possible to directly derive an (inefficient) quantum circuit that encodes the MPS[43,44]. To achieve this goal, one needs to consider each A_i as a rank-3 tensor, acting on vectors spanned by the left bond-index space, and transforming them into higher-dimension vectors spanned by the right bond-index and the physical index j_i. Note that the A_i matrices are defined up to a gauge transformation, which can be fixed by selecting a center of orthogonality and demanding that all tensors on its left and right sides, respectively satisfy $\sum A^{b_{i-1}b_i}_{j_i} A^{b'_{i-1}b_i\,*}_{j_i} = \delta^{b'_{i-1}}_{b_{i-1}}$ and $\sum A^{b_{i-1}b_i}_{j_i} A^{b_{i-1}b'_i\,*}_{j_i} = \delta^{b'_i}_{b_i}$. This gauge fixing procedure ensures that the tensors are injective and preserve orthogonality. Given the above gauge fixing, each tensor can be transformed into a unitary gate by adding an extra input index, while preserving its isometric property. This is analogous to adding columns to an orthogonal rectangular matrix until it becomes a square unitary matrix. By combining the resulting N gates in sequence, one obtains an exact circuit that creates the desired MPS. The problem with this encoding method, however, is that each of the encoder's gate acts on multiple qubits- one physical qubit and $\log_2(\chi)$ ancilla qubits associated with the bond index. Expressing such a gate in terms of native 2-qubit gates requires a complex circuit with depth exponential in the number of ancilla qubits[9,45–48], which would be too noisy for near-term devices (See section "Optimization algorithm for encoding MPS states" for more details).

A natural question is whether it is possible to obtain an *approximate* description of the state using a smaller number of two-qubit gates. A solution to this problem was proposed by Ran[44], following a layer-by-layer approach: (i) The first layer is obtained by simply truncating the bond dimension of the MPS to $\chi = 2$. The truncation of the bond dimension is a common procedure, obtained by contracting and re-separating each neighbor pairs of tensors trough a singular-value decomposition (SVD) and keeping a few of the largest singular values, in our case, the largest two. As explained above, the resulting state can be easily translated into a sequence of N 2-qubit gates. (ii) The next layer is generated by applying the inverse of the first layer to the target state and truncating the resulting MPS once again to $\chi = 2$, and so on and so forth. This protocol generates a staircase circuit of 2-qubit gates acting on neighboring pairs of qubits only, which we denote by $\{G_i^l\}_{i=1,\ldots,N}^{l=1,\ldots,L}$, where L is the number of layers, see Fig. 1a. This circuit is suitable for all near-term quantum computers, including processors with limited connectivity such as superconducting circuits, and will be used throughout this work.

The key advantage of the layer-by-layer approach is that each step of the calculation can be performed while keeping the state in an MPS form and using tensor contractions. In this way, the complexity of computing each layer grows linearly with the number of qubits, in contrast to a naive calculation whose complexity grows exponentially with the number of qubits. Note that the classical computational complexity scales quadratically with the bond dimension, which grows by a factor of 2 for each additional layer, i.e. exponentially with the number of layers. Hence, the maximal number of layers is bounded by the classical resources available and is typically set to 20 or less. To achieve a larger number of layers, it is necessary to perform a truncation of the MPS representation of the resulting state to a fixed bond dimension $\chi_{\max}$, at the probable cost of deteriorating the overlap with the target state. At present, however, this is not an important limitation, because near-term devices are anyway limited to a small number of layers.

A more severe problem is that the layer-by-layer approach is not effective: after the first few layers, any additional layer gives a negligible improvement. To highlight this point, we probe the infidelity $I = 1 - F$, where $F = |f|^2$ is the fidelity, and f is the overlap of the encoded state with the target state $|\psi_{\text{target}}\rangle$,

$$f(\{G_i^l\}) = \left\langle \psi_{\text{target}} \middle| \prod_{l=1}^{L} \prod_{i=1}^{N} G_i^l \middle| 0 \right\rangle. \tag{2}$$

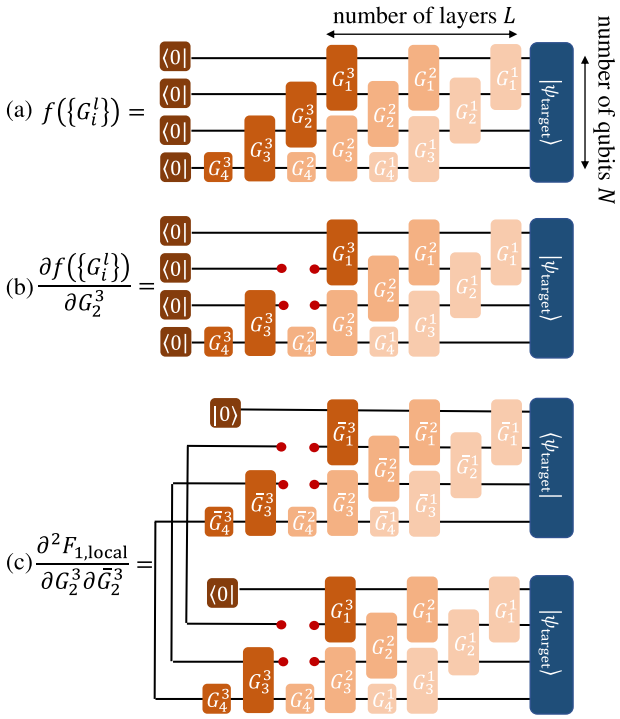


Fig. 1 | Tensor diagrams for the cost function and its derivatives. a Tensor diagram of the overlap between the target state $|\psi_{\text{target}}\rangle$ and a state encoded by a staircase circuit with $N = 4$ qubits and $L = 3$ layers. **b** Gradient of the overlap used in the global optimization protocol. **c** Second derivative of the local cost function used in optimization on quantum hardware. The red dots denote the free indices of the resulting tensor.

Fig. 2 shows in blue the infidelity of the layer-by-layer approach as a function of the number of layers, for two target states with $N = 12$ qubits and bond dimension $\chi = 64$ (see also Supplementary Figs. 1 and 2 for additional results with other system sizes, showing consistent results): (a) the ground state of an Ising model with the Hamiltonian $H = \sum_n J_z S_n^z S_{n+1}^z - h_x S_n^x$. This Hamiltonian is gapped for all $h_x \neq \pm J_z$. Its ground state is an example of a non-trivial state with low entanglement, which is faithfully described by MPS. Here, we consider a ferromagnetic state with $h_x/J_z = 0.6$; (b) a random MPS (See Methods section for more details). In both cases, the layer-by-layer approach shows a very slow convergence to the target state, as a function of the number of layers L. In fact, it is not even clear that the layer-by-layer algorithm will eventually converge to $I \to 0$ as $L \to \infty$. Moreover, we observe that the layer-by-layer approach is not capable of reproducing the correct circuit, even in cases where the target state is created using a small number of layers only. This is due to the random kernel used in the algorithm and to its inability to "plan ahead" and create cooperation between different layers. In each step, the algorithm maximizes only the immediate overlap of the state, without taking other amplitudes into account, which may potentially facilitate the performance of future (deeper) layers.

Results

Optimization algorithm for encoding MPS states

To improve the layer-by-layer approach, we introduce an optimization protocol that aims at finding the best circuit for a fixed number of layers and a fixed circuit topology. Developing a deterministic optimization algorithm requires us to evaluate the gradient of the cost function with respect to the

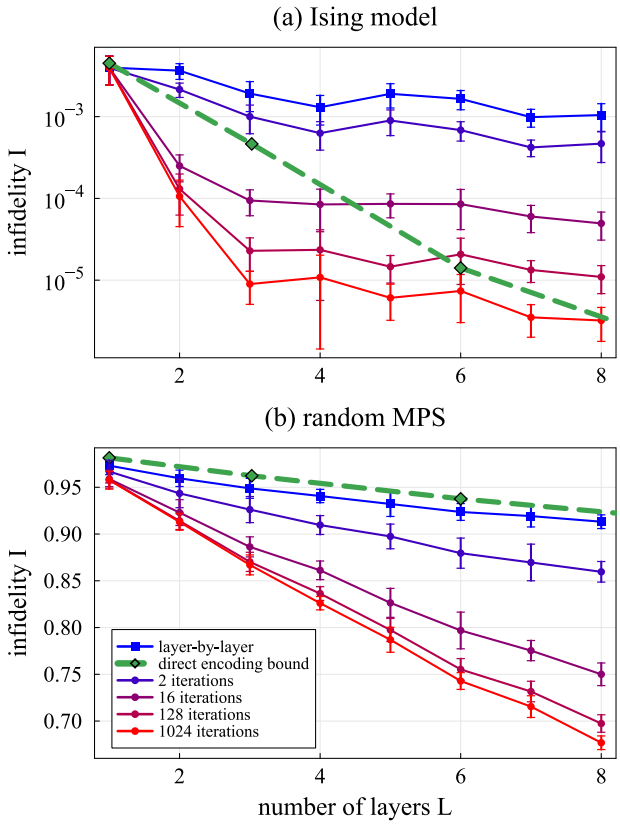


Fig. 2 | The infidelity *I* as a function of the number of layers *L*. The results are compared for two target states with $N = 12$ qubits and bond dimension of $\chi = 64$: **a** low entanglement ground state of the Ising model; **b** random MPS state. The obtained infidelity for different number of iterations is compared to the layer-by-layer approach[44], and to the direct bound given in the main text. The error-bars present the standard deviation of the infideility due to random circuit initializations and target states.

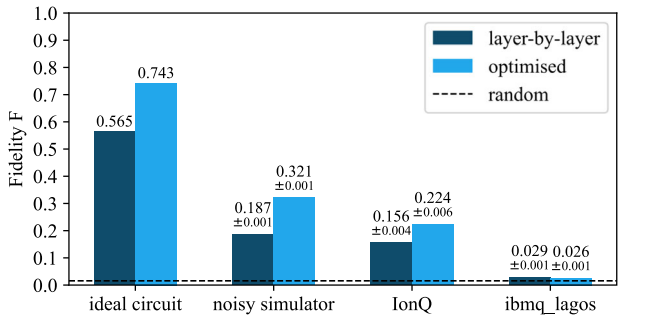


Fig. 3 | Fidelity of an encoded random state on different simulators and quantum computers. Fidelity of encoding a random-MPS target state of $N = 6$ qubits using circuits with $L = 3$ layers, obtained by the layer-by-layer method of ref. 44 and by the present optimization algorithm.

variational parameters, G_i^l. Fortunately, the gradient of the overlap f can be easily calculated using the linearity of the overlap with respect to a single gate. Taking the derivative of Eq. (2) removes one tensor from the contraction chain and leaves all other tensors, which act as a constant linear prefactor of the gate, intact. The final result is simply given by the original tensor network with one missing tensor, as illustrated in Fig. 1b. The gradient of f can, therefore, be efficiently contracted into a single rank-4 tensor with $2^4 = 16$ elements, using tensor-network methods. This gradient does not take into account the unitarity requirement of the gate G_i^l. To perform a gradient-descent optimization algorithm one has to follow the part of the gradient which preserves unitarity, see details in Supplementary Note I and in refs. 49–51. We, next, perform an element-by-element optimization scheme. This method, which has been used at first for optimization of sequentially generated entangled multi-qubit states and matrix product operators (MPOs) in refs. 52,53, is known also as the Evenbly-Vidal method for tensor network renormalization[54]. It has been since used in the context of variational quantum algorithms[55], quantum simulations[56], and tensor network optimization[57]. The method is analogous to the coordinate-descent method in classical machine learning[58] and to the iterative update of tensors in the DMRG algorithm[33] in the effort to optimize the objective function by parts. Unlike steepest descent methods, the element-by-element optimization does not rely on a convergence rate which needs continuous adjustment using heuristics, but rather proceeds through well defined and discrete steps. At each step, we select a single gate, G_i^l, and find the unitary matrix that maximizes $\mathrm{Tr}[(\partial f/\partial G_i^l)G_i^l]$. This matrix can be obtained by computing the tensor $\partial f/\partial G_i$ and projecting it to the closest unitary matrix. This last step simply involves the calculation of the SVD decomposition of the gradient and the substitution of the SVD values by 1. We, then, substitute the gate with the new optimum and proceed to the next gate. When all gates have been updated, the algorithm moves to the next iteration and runs again the protocol over all gates. The algorithm is repeated several times, until a specific convergence criterion is reached (see more details in Supplementary Note II).

As shown in Fig. 2, the proposed optimization protocol leads to a significant improvement over the layer-by-layer approach: (a) For the ground state of the Ising model, we achieve an improvement of up to one order of magnitude with just 10 iterations. We observe that the improvement is roughly inversely proportional to the number of iterations (the curves are equally spaced on a logarithmic scale). Hence, our protocol is efficient, in the sense that it can deliver infidelities arbitrarily close to the optimal value in a polynomial time. (b) For random states, the variational circuit is not expressive enough to achieve low values of the infidelity. Nevertheless, we observe that our optimization algorithm achieves a significant improvement over the layer-by-layer scheme and converges after a small number of iterations.

To check the efficiency of our method, we compare it to a more direct approach consisting of, first, reducing the size of the MPS tensors to a truncated bond dimension χ_0, and then, encoding the resulting state using k-qubit gates with $k = \log_2\chi_0 + 1$, using the method explained above. Such k-qubit gate can then be realized using $\Omega(k) = \frac{1}{9}4^k - \frac{1}{3}k - \frac{1}{9}$ 2-qubit gates[9]. Hence, using the direct approach requires the same number of 2-qubit gates as the optimized circuit with $L = \frac{4}{9}\chi_0^2 - \frac{1}{3}\log_2(\chi_0) - \frac{4}{9}$ layers, ignoring the reduced number of gates at the edges of the system. In particular, the case of $\chi_0 = 2$ is equivalent to a single layer of 2-qubit gates, $L = 1$, while $\chi_0 = 4$ corresponds to 3-qubit gates and requires $L = 6$ layers.

In Fig. 2 we compare the infidelities obtained using the direct method (dashed line) with those obtained by our optimization method (continuous lines). For the Ising model ground state, we find that at low depth the optimized infidelity is significantly lower than the infidelity of the direct encoding, while at higher depths the direct encoding is nearly optimal. In contrast, for the random MPS, the direct truncation method performs poorly in comparison to the optimized circuits. Our optimized circuit leads to better results because it takes into account higher singular values of the MPS which the truncation omits: when the tail of eigenvalues is significant, our approach performs much better than the direct encoding one.

Our optimization algorithm works under the assumption of ideal 2-qubit gates and an important question is what happens in a real noisy device. Due to the limited capabilities of available quantum processors, we considered the minimal instance of a target state where the proposed optimized circuit is expected to achieve a significant advantage over the layer-by-layer method, namely a random-MPS state with $N = 6$ qubits, encoded with $L = 3$ layers, see Methods section for details. Figure 3 compares the ideal fidelity with the results obtained in a noisy simulator and on quantum computers by IonQ and IBM. The circuit obtained by our optimization method outperformed the layer-by-layer encoding, except in the

case of the IBM Lagos quantum processor, whose faulty result is comparable to the fidelity of a random state, $F = 1/2^N$. As discussed earlier, the difference between the two encoding strategies is expected to grow with the accessible circuit depth.

State encoding on a quantum computer

Up to this point, we have assumed that the target state is known classically and all matrix elements can be computed using efficient tensor networks. We now move to a situation where the target state is not known classically, but can be prepared on a quantum computer. Encoding a state directly on a quantum computer is a desired technique for several reasons: For example, this scenario is common in the study of nonequilibrium quantum effects, such as quantum quenches where one aims at studying the time evolution of an initial state[5–8,56]. In addition, this approach is useful for the compilation of the initial layers of quantum machine learning algorithms[2,3,59,60]. Finally, it enables one to store the output of complex quantum calculations by obtaining a faithful shallow-circuit representation[61,62].

In all these applications, the initial step is to encode the target state using a small number of gates. Here we assume that a preexisting faithful copy of the desired state exists in the quantum computer and we search for an efficient encoder that is able to prepare the state using fewer number of gates. Without this assumption, training the ansatz on a quantum computer might not be scalable, as a cost function that is based on comparisons to tomography measurements in multiple bases will likely result in barren plateaus, due to its global nature. In a real quantum processor, the tensor derivative of the circuit cannot be performed analytically, but needs to be evaluated using tensor tomography. This process, which we refer to as gradient tomography, can be performed by measuring all possible products of Pauli matrices acting on the four sites in red in Fig. 1b (See Supplementary Note III for more details). Alternatively, the tomography can be performed using parameterization-dependant methods such as[63–66], which suggest a method for extracting the cost function dependency on a single rotation angle at a time. Performing full tomography using this method, however, would require resources exponential in the number of rotation angles, compared to polynomial resources with the Pauli-string substitution method. Due to the finite number of shots used to perform the tomography, the result of this procedure will include random fluctuations. For a typical number of shots of the order of a few thousands, the resulting uncertainties are of the order of few percents. The optimization algorithm is expected to work only as long as the gradients are large enough for the first steps of the optimization process.

Under generic conditions, the overlap between the initial configuration of the variational circuit and the target state decreases exponentially with the number of qubits. As a consequence, the amplitude of the gradient is exponentially suppressed and may not be detected experimentally. This situation, often referred to as barren plateaus[67–69], is exemplified in Fig. 4a for the case of random MPS target states with bond dimension $\chi = 64$, encoded using $L = 5$ layers. The figure displays the magnitude of the unitary gradient after the initialization of the circuit ansatz, at the beginning of the optimization algorithm. In the case of a random initialization, the gradient amplitude vanishes exponentially with the number of qubits, and follows the same slope as the average fidelity between random states, $F \sim 1/2^N$.

If the state is known in its MPS representation, one can try to beat the barren plateau by using a better initialization of the unitary gates. A smart initialization may bring the system to the proximity of the global minimum of the cost function, where the gradients are more pronounced. In our case, we can obtain a better initialization using the layer-by-layer algorithm. Figure 4a shows the unitary gradient obtained by applying the layer-by-layer initialization algorithm to a single layer (orange) or to all 5 (green). Due to the initialization the gradient is greatly increased, but still vanishes exponentially $\sim 0.75^{-N}$. For $N > 15$ the gradient already reaches the threshold value of 10^{-2} (dashed line), beyond which shot noise becomes dominant and the optimization protocol is stuck in a barren plateau.

A common technique to address barren plateaus is to substitute the *global* cost function, I, with a *local* one. This approach was shown

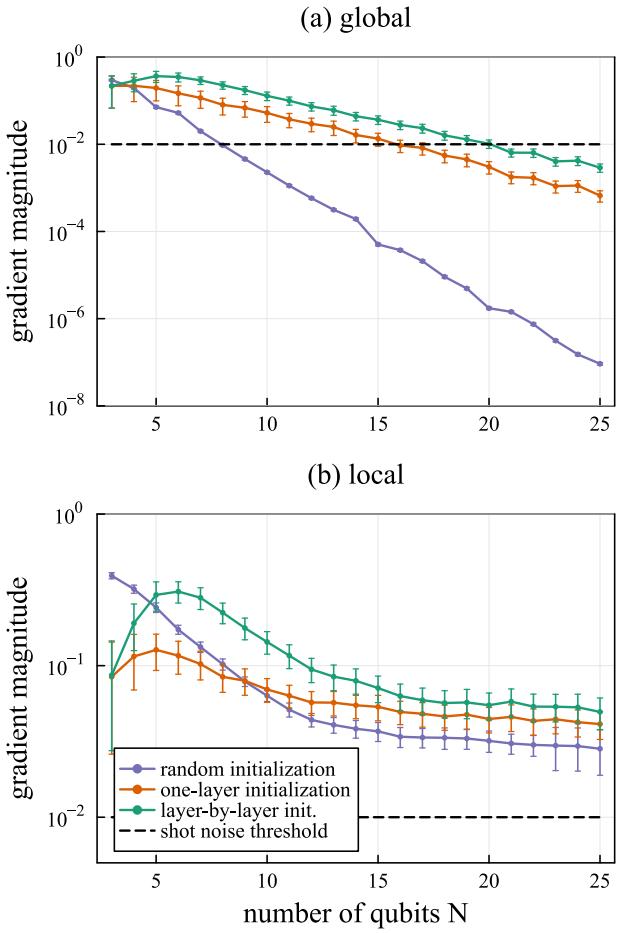


Fig. 4 | Magnitude of the unitary part of the initial gradient $\| \mathbf{grad}_G F \|$ for $L = 5$ layers, as a function of the number of qubits N. The gradient magnitude is simulated for different initalizations: **a** global cost function and **b** local cost function. In the global cost function **a**, the gradient decreases exponentially with N for all initializations, while in the case of a local cost function, it is polynomial. The dashed line represents the threshold for resilience to realistic shot noise (see text).

mathematically to solve the barren plateaus problem for shallow quantum circuit, i.e. for circuits where the circuit depth does not grow with the number of qubits[68]. Note that the mathematical theorems assumed a brick-wall gate structure, which is different than the present staircase gate connectivity. According to the definition of ref. 68, the present circuit has depth $L + N$ and, hence, beyond the regime of validity of the rigorous theorem. As we now show, our numerical calculations give evidence that the local cost function approach is valid in this case as well. We implement this strategy by introducing the local infidelity, $I_{\text{local}} = 1 - \sum_{n=1}^{N} F_{\text{local,n}}/N$, with

$$F_{\text{local,n}} = \left\langle \psi_{\text{target}} \right| \prod_{i,l} G_i^l P_n \prod_{i,l} \left(G_i^l\right)^\dagger \left| \psi_{\text{target}} \right\rangle \tag{3}$$

where $P_n = |0\rangle_n\langle 0|_n$ is the projection over the 0 state of the nth qubit. The fidelity $F_{\text{local,n}}$ represents the probability to measure the n'th qubit in the 0 state, irrespective of the state of the other qubits, and is maximal (equals to 1) when the circuit prepares the target MPS exactly. As shown in Fig. 4b, the usage of a local cost function solves the barren plateaus for our problem: the gradient is no longer an exponential function of the number of qubits and does not cross the shot-noise threshold up to a large number of qubits.

A potential drawback of the local cost function is that in the gate-by-gate optimization one needs to compute the derivative of quadratic functions $F_{\text{local,n}}$, rather than a linear one, f, see Eqs. (2) and (3). As a consequence, finding the optimal gate requires the measurement of a larger

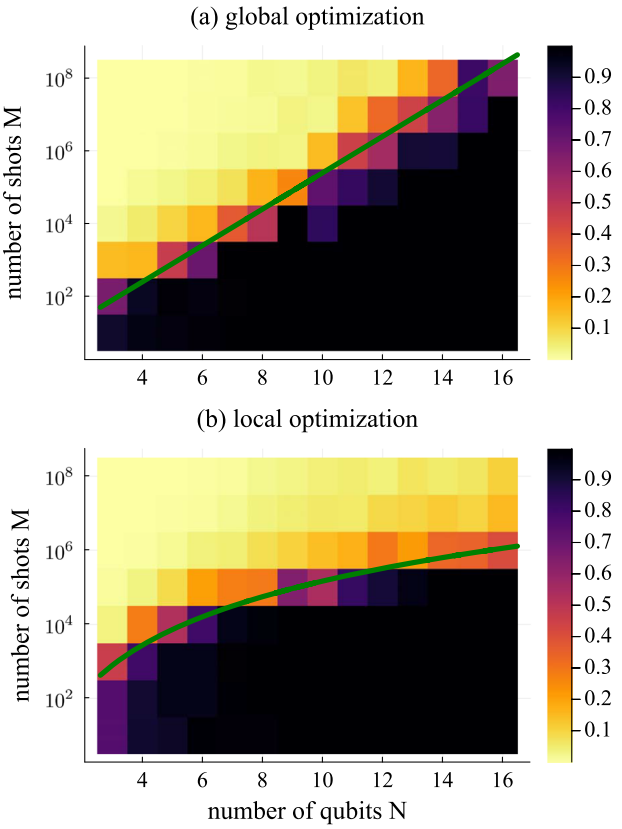


Fig. 5 | Infidelity *I* after circuit optimization as a function of the number of qubits and shots. The circuits was optimized using 20 iterations of **a** global and **b** local optimization algorithms, for a random-MPS target state with $L = 5$ layers and $\chi = 64$.

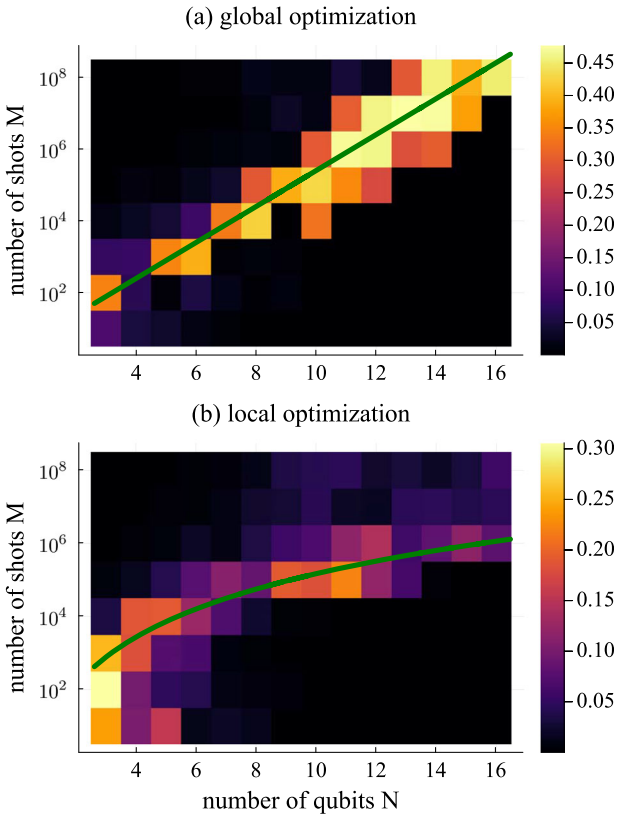


Fig. 6 | Standard deviation of the infidelity *I* as a function of the number of qubits and shots. Standard deviation of the infidelity for **a** global and **b** local optimization algorithms, for the same state and circuit presented in Fig. 5.

tensor of the second derivative as illustrated in Fig. 1c. Analytically, the second derivative of the local cost function is obtained by dropping the two copies of a gate G_i^l from the quadratic expression in (3), and is used to describe the full dependency of the quadratic local cost function on a single unitary gate. Measuring this tensor could potentially induce more errors in the calculation, and render it less effective for optimization. To estimate this effect, we test our optimization protocol on a phenomenological model that takes into account the finite number of shots per measurement, which we denote by M, see Methods section for details. Figure 5 compares the fidelity obtained by optimization using the global and local cost functions, for random-MPS target states. We find that, in the case of a small number of qubits $N \leq 8$, the local optimization is penalized by the larger number of measurements and leads to a larger infidelity. In contrast, for a large number of qubits, the local method performs better, as the polynomial scaling of the local gradient wins against the exponentially decreasing global gradient. The color coding (bright for small infidelities and dark for large infidelities) enables for a clear separation between successful and unsuccessful instances.

From the numerical data we can estimate the scaling of the minimal number of shots required by the global and local cost functions, by tracing the boundary between successful and unsuccessful optimization routines, a boundary highlighted in the variance the fidelity presented in Fig. 6, see green curves in Figs. 5 and 6. We obtain that for the global optimization, the minimal number of shots scales approximately as $N_{\text{shots}} \sim 3^N$, while for the local cost function it scales as $N_{\text{shots}} \sim N^4$, giving rise to an exponential improvement. These scaling laws can be directly translated into an estimation of the optimization runtime. For quantum computers based on superconducting transmon qubits, the time required for a single shot is of the order of 100 micro-seconds (see, for example, ref. 70). In such a system, for a 15 qubit state, the optimization of each single gate requires 10^9 shots, corresponding to an experimental runtime of more than 1 day. In contrast, using a local cost function the same task can be performed using only 10^6 shots, requiring less than 2 minutes.

Discussion

In this work, we addressed the problem of quantum state encoding, namely how to encode a target state using 1-qubit and 2-qubit gates only. We opened our study with MPSs, an efficient representation of quantum states with low levels of entanglement. Realizing these states on a quantum computer requires a circuit depth that grows at least linearly with the bond dimension χ. Here, we showed how to create a state that approximates the MPS using a smaller number of gates. Our approach focuses on quantum circuits with a fixed depth and aims at finding the gate configuration that offers the best approximation to the MPS. We described an optimization technique that iteratively improves each gate, until the optimal solution is found. Each step involves a singular value decomposition of a small tensor, whose elements can be computed efficiently using tensor network techniques and require limited resources only. By considering two extreme types of MPSs (low-entanglement states and random states), we demonstrated that our algorithm converges to a solution with a good fidelity. Empirically we observed that the algorithm is not hindered by local minima, unlike other variational approaches where local minima are abundant[27].

Next, we considered the realization of the proposed element-by-element optimization algorithm directly on a quantum computer. This task is relevant to situations where the MPS is not known classically, such as in the case of a quantum dynamics simulations, encoding the outputs of complex quantum calculations. In a quantum computer the tensors can be estimated up to a finite resolution only, due to the finite number of measurements. In this case, the optimization algorithm encounters barren plateaus. We explained how to solve this problem by moving to a local cost function, and demonstrated its performance by considering the effects of realistic shot noise.

The main outcome of our work is a generic algorithm that allows one to encode a quantum state using a small number of gates. Unlike gradient-based optimization algorithms, this algorithm is monotonic, in the sense that each step necessarily brings us closer to the final goal. Its key ingredient is the direct estimation of the tensor derivative of the circuit and its translation into an improved set of gates. An important question that deserves future investigation is how to choose the circuit depth and the gate connectivity that will deliver the largest fidelity. On the one hand, by considering a larger number of layers and a more complex connectivity, one expects to obtain a larger fidelity. On the other hand, in the real world, adding more gates leads to more decoherence and noise, and, hence, to a smaller fidelity. Finding the optimal number will generically depend on the entanglement of the target state and on the gate faultiness of the device. Studying this dependence theoretically and experimentally is a fundamental step towards demonstrating the applicability of our approach.

Methods

Generation of randomly initialized circuits and random MPS

Throughout this work, random circuits were generated using gates drawn from the Haar-random distribution. To obtain Haar-random unitary matrices we used the common prescription of generating element-wise random Gaussian matrices and replacing their singular values with ones.

Random MPS were generated using iterative evolution and truncation. We started with the zero state $|0\rangle^{\otimes N}$ as a seed and iteratively contracted it with a stair-case layer of unitary Haar-random gates, followed by truncation to a predefined maximal bond dimension. The process is repeated 1000 times to obtain a fully-randomized MPS with a bounded bond dimension.

Implementation on a real device

To test our method for MPS encoding on a real quantum processors we chose a quantum state for which the difference between the layer-by-layer encoder and the optimized encoder is significant. Due to the limited capability of state-of-the-art devices, we chose a small random state with $N = 6$ qubit and encoded it using $L = 3$ layers. We numerically sampled 1000 random states and calculated the fidelity of the layer-by-layer encoder for each one. We, then, chose the random state that exhibited the lowest fidelity and applied our optimization algorithm to obtain a better encoder. The encoder circuits were transpiled into a sequence of native gates of single qubit rotations and CNOT gates using Qiskit libraries, which implement an exact 2-qubit gate decomposition by a sequence of 3- CNOT gates interleaved with 1-qubit rotation gates, as detailed in[71]. The original (optimized) transpiled circuit contained 142 (163) 1-qubit rotations, and 23 (24) CNOT gates. Next, we performed full quantum state tomography of the output state, using $3^6 = 729$ circuits, with at least 200 measurement shots each. From these measurements, we estimated the density matrix of the final state, ρ_{exp}, and computed the fidelity to the target state as $F = \langle\psi_{\mathrm{MPS}}|\rho_{\mathrm{exp}}|\psi_{\mathrm{MPS}}\rangle$.

The noisy simulation was performed using Qiskit Aer with noise parameters extracted from the ibm_lagos quantum processor. As shown in Fig. 3, the actual device performed much worse and we attribute this discrepancy to the limited connectivity of the device: our algorithm assumes a linear connectivity between the 6 qubits, which is unavailable on the device. The Qiskit compiler solves this problem by adding SWAP gates between qubits that are not directly connected, but these extra gates lead to a much worse performance.

Realistic model of shot noise

Our optimization algorithm is based on the measurement of local gradients, obtained by disconnecting a single gate from the circuit. To estimate the effects of shot noise, we expanded the gradient tensors in sums of Pauli strings and, then, replaced the coefficient of each Pauli string by a random variable sampled from a binomial distribution with M trials. This substitution left the average value of the tensor unchanged and mimicked the result of a quantum measurement with M shots. Our model disregards the effects of phase noise, which may lead to a further suppression of the gradient, but is sufficient to obtain a qualitative estimate of the effects of shot noise on the gradients.

Data availability

All data and figures that support the findings of this paper are available from the corresponding author on request. Please refer to Matan Ben-Dov at matan.ben-dov@biu.ac.il.

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Author contributions

E.G.D.T. and A.M. conceived the idea for this work. E.G.D.T. and M.B.D. developed the theoretical framework detailed in this work, M.B.D. designed and performed the simulations and analyzed their results. Diagrams were

created by E.G.D.T. and figures were created by M.B.D.; D.S. performed and analyzed the experiments on quantum computers. M.B.D. and E.G.D.T. wrote the manuscript, and A.M. revised it.

Competing interests
The authors declare no competing interests.

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